Taxi PDP problem

Foreword

This program aims at simulating a Taxiing problem in a warehouse/on a map.

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Orders

Each order has random pickup and delivery points. Orders are defined late if delivered after a time window defined by: TW times the time it should take an agent to deliver the package using the shortest path from pickup to delivery.

Graph

The graph needs to be directed. The distances of the edges have to fit the vertices positions. The larger the loops, the better.

Agents

Each agent has its own processing unit. There are 3 main categories: AGVs and Packs

AGVs

They send ants carrying their intention to either pickup a Pack, or to tell other AGVs they are stuck and would like to move.

Each AGV tries to go to the closest Pack on the map, provided it is free. In other words, one unique AGV moves to get a given Pack.

They are able to negotiate with other agents and to detect and (theoretically) escape deadlocks. The process of negotiation is simple but does the job: When an AGV is stuck by another one, it gives its burden to the blocking AGV. If it is already carrying a Pack, then it just sends a move request with the list of all AGVs blocked in that queue.

A deadlock is discovered when a loop of move requests is detected or when an AGV is idle for more than 60 seconds. Then, it moves randomly to an available node and recomputes its path.

Packs

They send ants carrying their time window and the path to get there. They have to select which AGV has to pick them up. In this case, they also propagate the ID of the AGV they have booked.

Customers

Delivery points of the packs, they just send ants to propagate the path to their position.

Delegate Agents: Ants

Two types of ants: Feasibility ants and Intention ants. They move at an infinite speed.

Feasibility ants

Sent by Packs and Customers. They multiply and leave dropsigns at each node. Dropsigns carry all information useful for picking up/delivering a Pack.

Intention ants

Delegate agents of AGVs. They go to the position the AGV sent them to, then share desired information. They're in charge of booking nodes by means of Flags carrying the AGV's ID, of telling other AGVs they are blocking the way and of telling a Pack their AGV is a candidate for picking it up.

Running the code

The .jar file can be run with the provided file run1.sh. It contains detailed preset parameters.

We also included other files we used to generate our results in an automated way:

- runTests1.sh: main script running simulations with different parameters and post-processing;
- stats?.m: Matlab files generating plots;
- tabularizer.sh: script generating LaTeX tables from ascii data.